Chapter 5 — Lists, Tuples and Arrays

Python gives you several ways to keep items **in order**: the all-purpose `list`, the immutable `tuple`, the compact numeric `array.array`, and — from the scientific world — NumPy's `ndarray`. They look alike (all support indexing, slicing, `len`, `in`, and `for`) but behave very differently in mutability, memory, speed and safety. This chapter explains each one, how it works under the hood, and how to choose.

What you'll learn

1. The `array` module: type codes, indexing, insertion, deletion, search, update, and its methods.
2. Lists: creating, accessing, updating, joining, slicing, looping and operators.
3. How a list is really a dynamic array — and the big-O cost of every operation.
4. Adding and removing items, and the full list-method table.
5. Advanced slicing: steps, slice assignment, slice deletion.
6. List comprehensions.
7. Sorting: `sort()` vs `sorted()`, `key`, `reverse`, stability, multi-key sorts.
8. Copying: aliasing, shallow vs deep copies, and the `[[0]*3]*3` trap.
9. Tuples: immutability, operators, unpacking (including `*`), joining, methods, tuples as records, `namedtuple`.
10. When to choose `list`, `tuple`, `array.array` or NumPy.

All runnable code is in `code/`. Outputs below are real, produced with Python 3.14 (the code runs on 3.12+). Timings are from one laptop — yours will differ, but the ratios hold.

5.1 The `array` module — typed, compact arrays

What it is

An **array** (from the standard-library `array` module) is a mutable sequence in which **every element has the same C numeric type** — all 32-bit ints, or all 64-bit floats, and so on. The values are stored as raw machine numbers side by side in one block of memory (**contiguous** storage), rather than as separate Python objects.

Analogy: a list is a shelf of labelled boxes, each box holding any object; an `array` is an egg carton — every slot is the same size and holds one kind of thing, so it packs far more densely.

Why it matters

For large volumes of plain numbers — sensor readings, audio samples, pixel values, binary file formats — an `array` uses a fraction of a list's memory and can be written to or read from files and sockets directly as bytes.

How it works

1. `import array` and create one with a **type code** that fixes the element type: `array.array("i", [10, 20, 30])`.
2. Every value you store is checked and converted to that C type; wrong types raise `TypeError`, out-of-range numbers raise `OverflowError`.
3. Indexing (positive and negative), slicing, `len`, `in` and `for` work exactly as with lists.

Type codes (sizes are the minimum the C standard guarantees; `itemsize` tells you the real size on your machine)

Code	C type	Python type	Min bytes	Typical on 64-bit macOS/Linux
`'b'` / `'B'`	signed / unsigned char	int	1	1 (range −128…127 / 0…255)
`'h'` / `'H'`	signed / unsigned short	int	2	2
`'i'` / `'I'`	signed / unsigned int	int	2	4
`'l'` / `'L'`	signed / unsigned long	int	4	8 (4 on Windows)
`'q'` / `'Q'`	signed / unsigned long long	int	8	8
`'f'`	float	float	4	4
`'d'`	double	float	8	8
`'u'`	wchar_t (Unicode char)	str	2	4 – **deprecated** since 3.3, slated for removal
`'w'`	Py_UCS4 (Unicode char)	str	4	4 – added in 3.13 as the replacement for `'u'`

```
a = array.array("i", [10, 20, 30, 40, 50])      # 'i' = C signed int
print(a, a.typecode, a.itemsize, "bytes per item")
print(a[0], a[1], a[-1], a[-2])
for x in a:
    print(x, end=" ")
try:
    a.append(3.5)                               # wrong type is rejected
except TypeError as e:
    print("TypeError:", e)
try:
    array.array("b", [200])                     # 'b' holds -128..127
except OverflowError as e:
    print("OverflowError:", e)
```

Output:

```
array('i', [10, 20, 30, 40, 50]) i 4 bytes per item
10 20 50 40
10 20 30 40 50
TypeError: 'float' object cannot be interpreted as an integer
OverflowError: signed char is greater than maximum
```

The four basic operations: insert, delete, search, update

Operation	Code	Effect	Cost
Traverse	`for x in a`	Visit each element	O(n)
Insert	`a.insert(i, x)`	Put `x` at index `i`, shift the rest right	O(n)
Delete	`a.pop(i)` / `a.remove(x)`	Remove by position / first matching value, shift left	O(n) (O(1) for `pop()` at the end)
Search	`a.index(x)`	Position of the first `x` (or `ValueError`)	O(n)
Update	`a[i] = x`	Overwrite in place	O(1)

Array methods

Method	What it does	Cost
`append(x)`	Add `x` at the end	O(1) amortised
`extend(iterable)`	Append every item of an iterable (same type required)	O(k)
`insert(i, x)`	Insert before position `i`	O(n)
`pop([i])`	Remove and return item `i` (default last)	O(1) last, O(n) otherwise
`remove(x)`	Delete the first occurrence of `x`	O(n)
`index(x)`	Position of first `x`	O(n)
`count(x)`	How many times `x` occurs	O(n)
`reverse()`	Reverse in place	O(n)
`clear()`	Remove all items (Python 3.13+; use `del a[:]` on older versions)	O(1)
`tolist()` / `fromlist(l)`	Convert to / append from a list	O(n)
`tobytes()` / `frombytes(b)`	Raw machine bytes out / in	O(n)
`tofile(f)` / `fromfile(f, n)`	Write / read raw values to a binary file	O(n)
`byteswap()`	Swap byte order (endianness) of every item	O(n)
`buffer_info()`	`(address, length)` of the underlying memory	O(1)

Two things arrays **don't** have: a `.sort()` method and a `.copy()` method. Sort with the built-in `sorted(a)` — which returns a **list** — and convert back with `array.array(code, sorted(a))`. Copy with `a[:]`, `array.array(a.typecode, a)` or `copy.copy(a)`.

```
a = array.array("i", [1, 2, 3, 4, 5])
a.insert(1, 10);            print("insert(1, 10):", a.tolist())
removed = a.pop(1);         print("pop(1):", a.tolist(), "removed", removed)
print("index(3):", a.index(3))
a[2] = 20;                  print("a[2] = 20:", a.tolist())
a.append(4);                print("append(4):", a.tolist())
a.extend([5, 4, 6]);        print("extend:", a.tolist())
print("count(4):", a.count(4))
a.remove(4);                print("remove(4) (first 4 only):", a.tolist())
a.reverse();                print("reverse:", a.tolist())
print("sorted(a) -> list:", sorted(a))          # arrays have no .sort()
a = array.array("i", sorted(a));  print("back to array:", a)
b = copy.copy(a)                                # no .copy() method: use copy.copy or a[:]
b[0] = 99
print("copy independent:", a[0], b[0])
del a[:]                                        # empties it (array.clear() exists from 3.13)
print("emptied:", a)
```

Output:

```
insert(1, 10): [1, 10, 2, 3, 4, 5]
pop(1): [1, 2, 3, 4, 5] removed 10
index(3): 2
a[2] = 20: [1, 2, 20, 4, 5]
append(4): [1, 2, 20, 4, 5, 4]
extend: [1, 2, 20, 4, 5, 4, 5, 4, 6]
count(4): 3
remove(4) (first 4 only): [1, 2, 20, 5, 4, 5, 4, 6]
reverse: [6, 4, 5, 4, 5, 20, 2, 1]
sorted(a) -> list: [1, 2, 4, 4, 5, 5, 6, 20]
back to array: array('i', [1, 2, 4, 4, 5, 5, 6, 20])
copy independent: 1 99
emptied: array('i')
```

`tolist()` is used only to print a clean list; printing the array itself shows `array('i', [...])`.

Memory. One million integers:

```
list of 1,000,000 ints :   36.0 MB (pointers + int objects)
array('i')             :    4.1 MB (raw 4-byte ints)
```

A list stores an 8-byte pointer per slot **plus** a separate int object (28 bytes each for small ints on 64-bit CPython) — about 9× more memory here.

Trade-offs, mistakes, and when not to use it

- **Not faster for arithmetic.** Each time you read an element, Python must create an `int` / `float` object for it ("boxing"), so looping over an array is often *slower* than over a list (see 5.15). For fast maths, use NumPy.
- **Choose the type code carefully:** `'b'` overflows at 128; `'l'` is 4 bytes on Windows but 8 on Linux/macOS — use `'q'` when you need 64 bits everywhere.
- Use arrays for **compact storage and binary I/O**; use lists for general work; use NumPy for numerical computing.

Self-check

1. What does the type code decide? → The C type, hence size and allowed values, of every element.
2. How do you sort an `array`? → `array.array(a.typecode, sorted(a))`.
3. Why does an array of a million ints use ~4 MB while a list uses ~36 MB? → The array stores raw 4-byte values; the list stores pointers to full Python int objects.

TL;DR

`array.array(code, data)` = homogeneous, compact, mutable numbers. Same indexing and most list methods; no `sort` / `copy` methods; `clear()` only from 3.13. Great for memory and binary I/O, not for speed.

5.2 Lists — the everyday sequence

What it is

A **list** is an ordered, **mutable** collection that can hold objects of **any type**, written with square brackets: `[1, "Hello", 3.14, True]`. It can grow and shrink at run time.

Analogy: a to-do list on a whiteboard — you can read any line, rewrite it, add lines, erase lines, and reorder them.

Why it matters

It's the default container in Python. Function results, rows from a file, items from an API, stacks, queues of work — you'll reach for a list first in most programs.

How it works — create, access, update

```
my_list = [1, "Hello", 3.14, True]          # mixed types are fine
print(my_list, len(my_list))
print(my_list[0], my_list[1], my_list[-1], my_list[-2])
my_list[1] = "World"
print("single update:", my_list)
my_list[1:3] = ["Changed", "Completely"]
print("slice update:", my_list)
print(list("abc"), list(range(4)), [0] * 3, [])
```

Output:

```
[1, 'Hello', 3.14, True] 4
1 Hello True 3.14
single update: [1, 'World', 3.14, True]
slice update: [1, 'Changed', 'Completely', True]
['a', 'b', 'c'] [0, 1, 2, 3] [0, 0, 0] []
```

- Indexing works like strings: `0` is first, `-1` last, `-2` second-last.
- Unlike strings, you can **assign** to an index (`my_list[1] = ...`) or a whole slice (`my_list[1:3] = [...]`) — see 5.5.
- `list(iterable)` builds a list from anything iterable.

How it works — joining, slicing, looping

```
a, b = [1, 2, 3], [4, 5, 6]
c = a + b                                   # new list
a.extend(b)                                 # modifies a in place
print(c, a, c == a, c is a)
full = list(range(10))
print(full[2:7], full[-8:-3], full[::3], full[::-1])
for i, fruit in enumerate(["apple", "banana", "cherry"], start=1):
    print(i, fruit)
```

Output:

```
[1, 2, 3, 4, 5, 6] [1, 2, 3, 4, 5, 6] True False
[2, 3, 4, 5, 6] [2, 3, 4, 5, 6] [0, 3, 6, 9] [9, 8, 7, 6, 5, 4, 3, 2, 1, 0]
1 apple
2 banana
3 cherry
```

`+` builds a **new** list and leaves both inputs alone; `extend()` (or `+=`) **modifies** the left list in place. `c == a` compares contents (equal); `c is a` compares identity (different objects).

List operators

Operator	Meaning	Example	Result
`+`	Concatenate into a new list	`[1, 2] + [3]`	`[1, 2, 3]`
`*`	Repeat	`[1, 2] * 2`	`[1, 2, 1, 2]`
`in` / `not in`	Membership (linear scan)	`3 in [1, 2, 3]`	`True`
`==`, `<`, …	Element-by-element (lexicographic) comparison	`[1, 2, 3] < [1, 3]`	`True`
`+=`	Extend in place	`a += [4]`	`a` changed
`del a[i]`	Delete by index or slice	`del a[0]`	

```
a, b = [1, 2, 3], [4, 5, 6]
print(a + b, a * 3, 3 in a, 7 not in a)
print([1, 2, 3] < [1, 3], [1, 2] == [1, 2], max(a), min(a), sum(a))
```

Output:

```
[1, 2, 3, 4, 5, 6] [1, 2, 3, 1, 2, 3, 1, 2, 3] True True
True True 3 1 6
```

`[1, 2, 3] < [1, 3]` is `True` because the first elements tie (1 = 1) and at the second position 2 < 3 — exactly like comparing words letter by letter.

Common mistakes

- Confusing `a + b` (new list) with `a.extend(b)` (in place, returns `None`). Writing `a = a.extend(b)` sets `a` to `None`.
- Testing membership in a big list inside a loop — each `in` is O(n). Convert to a `set` first (Chapter 6).
- Modifying a list while iterating over it (e.g. removing items) — elements get skipped. Iterate over a copy (`for x in lst[:]`) or build a new list with a comprehension.

Self-check

1. `[1, 2] * 3`? → `[1, 2, 1, 2, 1, 2]`.
2. What does `a.extend(b)` return? → `None` (it changes `a`).
3. `full[-8:-3]` on `list(range(10))`? → `[2, 3, 4, 5, 6]`.

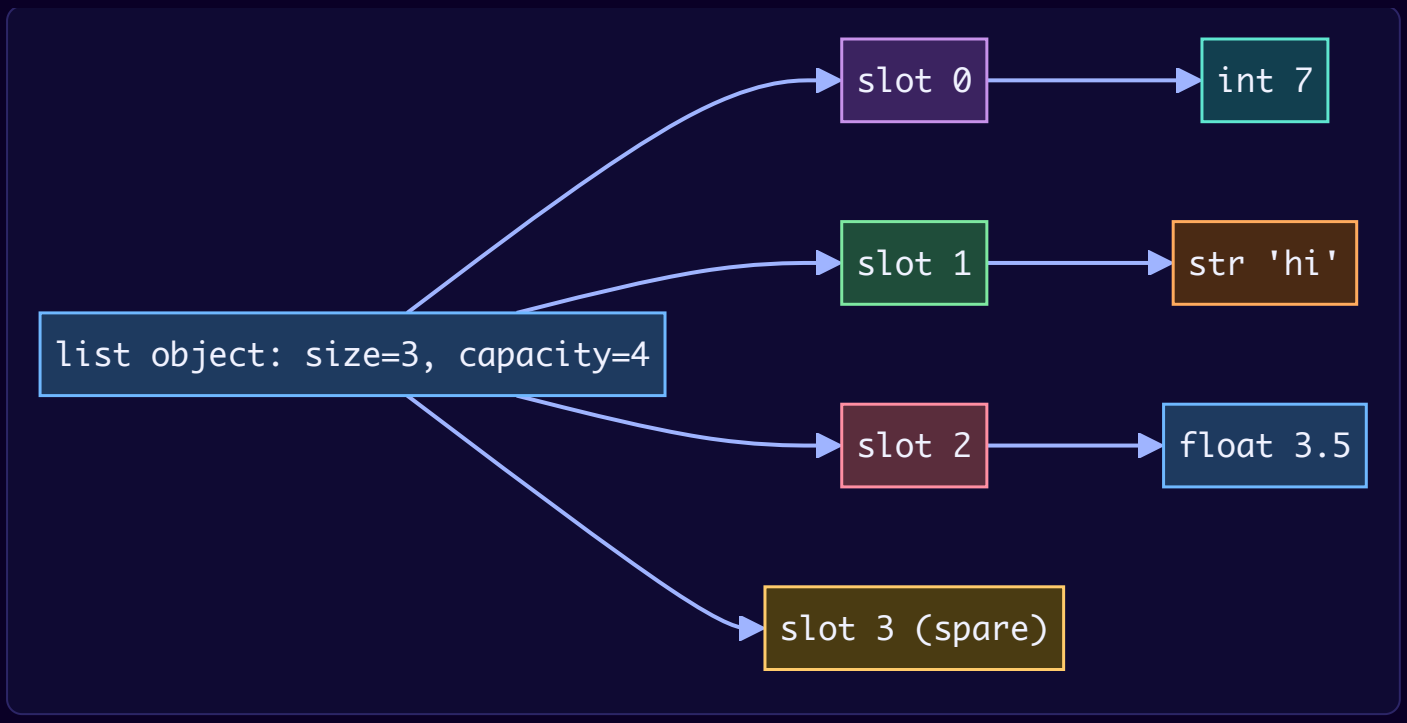
TL;DR

Lists are ordered, mutable, heterogeneous. Index, slice, update in place; `+` makes new lists, `extend`/`+=` modify.

5.3 List internals — a dynamic array — and the big-O table

What it is

In CPython a list is a **dynamic array**: one contiguous block of memory holding **pointers** (references) to the element objects, plus two numbers — how many slots are used (`len`) and how many are allocated (**capacity**). "Dynamic" means it resizes itself automatically.



Why it matters

Knowing the layout tells you instantly which operations are cheap and which are slow — the difference between an O(n) and an O(n²) program.

How it works

1. **Index access** `lst[i]`: address = start + i × 8 bytes → one step, O(1).
2. **Append** when there is spare capacity: write the pointer into the next slot, O(1).
3. **Append when full**: allocate a bigger block, copy all pointers, then write. CPython grows by roughly **12.5% plus a small constant** (the formula in recent CPython versions is `new_size + new_size/8 + 6`, rounded down to a multiple of 4). Because each resize makes room for many future appends, the copying cost spread over all appends is constant — **amortised O(1)**.
4. **Insert or delete at position i**: every pointer after `i` must shift one slot → O(n − i). At the front (`i = 0`) that's the whole list.
5. **Shrinking**: when the list falls below half its capacity, CPython may reallocate a smaller block.

Measured capacity while appending (from `code/06_list_internals.py`, using `sys.getsizeof`):

```
len  capacity  (capacity = slots allocated)
  0         0
  1         4
  5         8
  9        16
 17        24
 25        32
 33        40
 41        52
 53        64
 65        76
```

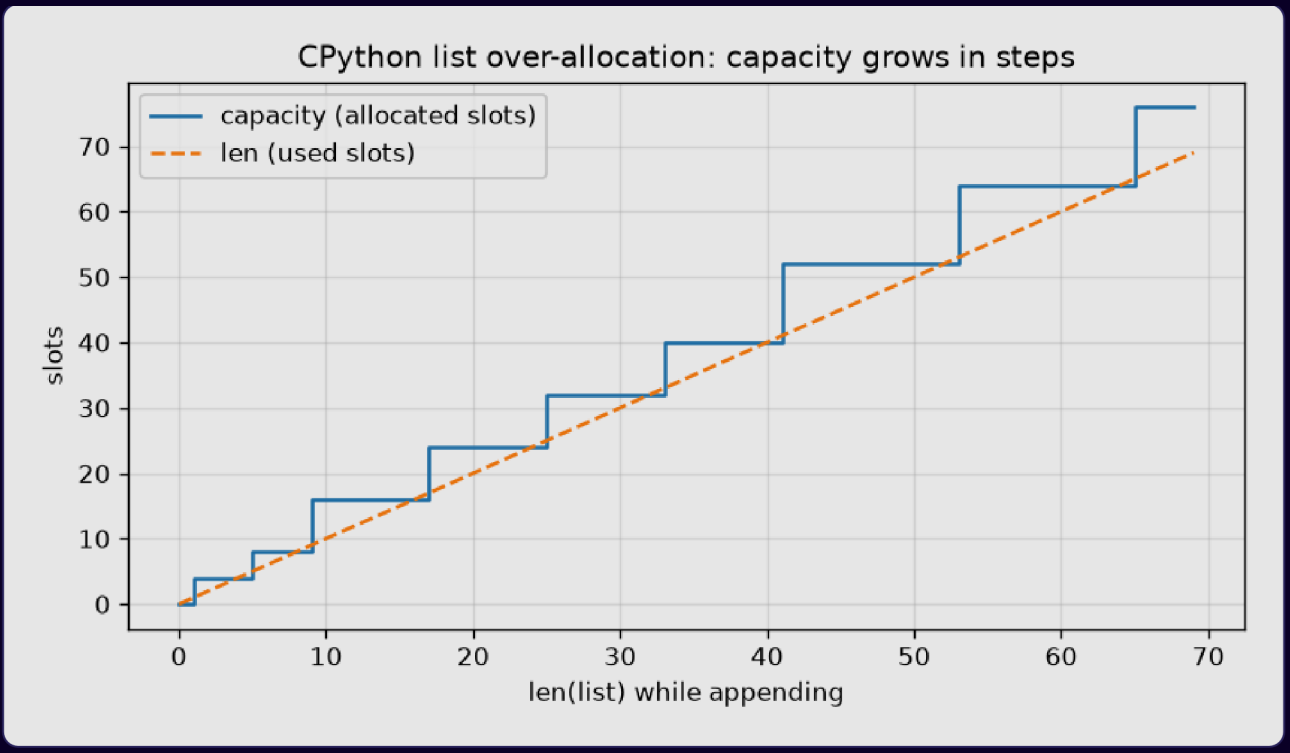


Figure — capacity (steps) always stays at or above the used length (dashed line); a copy happens only at each step.

Worked amortised example: appending the 17th item finds the list full at 16, so CPython allocates 24 slots and copies 16 pointers. The next 7 appends are free. Spread over those 8 appends, that's 2 copies each — and as lists grow, the steps get larger in absolute terms, so the average copy cost per append stays bounded by a constant.

Timing the difference between the end and the front (µs per push-then-pop pair):

n	append()+pop()	insert(0)+pop(0)	(µs per pair)
10,000	0.017	3.588	
100,000	0.017	40.046	
1,000,000	0.018	414.395	

End operations stay flat as *n* grows ×100; front operations grow ×100 — the signature of O(1) vs O(n).

Big-O table for `list` (n = len, k = size of the other argument or slice)

Operation	Example	Average cost
Length	`len(a)`	O(1)
Get / set item	`a[i]`, `a[i] = x`	O(1)
Append	`a.append(x)`	O(1) amortised
Pop last	`a.pop()`	O(1)
Pop / delete at i	`a.pop(i)`, `del a[i]`	O(n − i) → O(n) at the front
Insert at i	`a.insert(i, x)`	O(n − i)
Extend	`a.extend(b)`, `a += b`	O(k)
Concatenate	`a + b`	O(n + k)
Repeat	`a * k`	O(n·k)
Get slice	`a[i:j]`	O(k)
Set slice	`a[i:j] = b`	O(k + n)
Delete slice	`del a[i:j]`	O(n)
Membership / search	`x in a`, `a.index(x)`, `a.count(x)`, `a.remove(x)`	O(n)
Min / max / sum	`min(a)`	O(n)
Copy	`a.copy()`, `a[:]`	O(n)
Reverse	`a.reverse()`	O(n)
Sort	`a.sort()`, `sorted(a)`	O(n log n) worst, O(n) if already sorted
Iterate	`for x in a`	O(n)

Trade-offs

Need	Best structure	Why
Append/pop at the end (stack)	`list`	O(1)
Add/remove at both ends (queue)	`collections.deque` (Chapter 6)	O(1) at both ends; list's `pop(0)` is O(n)
Frequent membership tests	`set` / `dict`	O(1) average instead of O(n)
Keep sorted while inserting	`bisect` on a list (Chapter 6) or a sorted container library	O(log n) search, O(n) insert

Common mistakes

- Using `list.pop(0)` as a queue — quadratic for large queues. Use `deque.popleft()`.
- Calling `insert(0, x)` in a loop to build a reversed list — append then `reverse()` once.

In the interview

> **30-second answer:** "A Python list is a dynamic array of pointers. Indexing and append are O(1) (append amortised, thanks to proportional over-allocation of about 12.5%), but inserting or deleting anywhere except the end shifts elements and is O(n), and membership tests are O(n)."

- *Q: Why amortised?* A: Occasional resizes copy everything, but they happen rarely enough that the total copying over n appends is O(n).
- *Q: Linked list vs Python list?* A: Python has no built-in linked list; `deque` is a doubly-linked list of fixed-size blocks, giving O(1) at both ends but O(n) indexing in the middle.
- *Q: Why is `x in lst` slow?* A: It compares against each element in turn — linear scan.

Self-check

1. What is stored in a list's slots? → Pointers to objects, not the objects themselves.
2. Cost of `lst.insert(0, x)` on a million-item list? → O(n): a million pointers move.
3. Why doesn't every append copy the list? → Spare capacity is reserved in advance.

TL;DR

List = dynamic array of pointers. O(1): index, append, pop-last. O(n): insert/delete at front or middle, `in`, `index`, `remove`. Queue → `deque`; lookups → `set`.

5.4 Adding and removing items

What it is

Methods and statements that change a list's contents.

Goal	Use	Returns	Error when
Add one item at the end	`append(x)`	`None`	–
Add one item at position i	`insert(i, x)`	`None`	– (i beyond ends is clamped)
Add all items of an iterable	`extend(iterable)` / `+=`	`None`	Not iterable
Remove first item equal to x	`remove(x)`	`None`	`ValueError` if absent
Remove by position and get it	`pop(i=-1)`	The item	`IndexError` on empty/out of range
Remove by position or slice	`del a[i]`, `del a[i:j]`	–	`IndexError`
Remove everything	`clear()` (or `del a[:]`)	`None`	–

```
fruits = ["apple", "banana"]
fruits.append("cherry");              print("append:", fruits)
fruits.insert(1, "orange");           print("insert:", fruits)
fruits.extend(["grape", "mango"]);    print("extend:", fruits)
fruits.remove("banana");              print("remove:", fruits)
last = fruits.pop();                  print("pop():", fruits, "->", last)
first = fruits.pop(0);                print("pop(0):", fruits, "->", first)
del fruits[0];                        print("del [0]:", fruits)
fruits.clear();                       print("clear:", fruits)
x = [1, 2]
x.append([3, 4]);  print("append a list :", x)   # nests!
y = [1, 2]
y.extend([3, 4]);  print("extend a list :", y)
try:
    y.remove(99)
except ValueError as e:
    print("ValueError:", e)
```

Output:

```
append: ['apple', 'banana', 'cherry']
insert: ['apple', 'orange', 'banana', 'cherry']
extend: ['apple', 'orange', 'banana', 'cherry', 'grape', 'mango']
remove: ['apple', 'orange', 'cherry', 'grape', 'mango']
pop(): ['apple', 'orange', 'cherry', 'grape'] -> mango
pop(0): ['orange', 'cherry', 'grape'] -> apple
del [0]: ['cherry', 'grape']
clear: []
append a list : [1, 2, [3, 4]]
extend a list : [1, 2, 3, 4]
ValueError: list.remove(x): x not in list
```

Common mistakes

- `append` vs `extend`: `append([3, 4])` adds **one** element (a nested list); `extend([3, 4])` adds **two**. `extend("hi")` adds `'h'` and `'i'`.
- Removing all occurrences with a `while x in lst: lst.remove(x)` loop — $O(n^2)$. Use `lst = [v for v in lst if v != x]`.
- `lst.clear()` vs `lst = []`: `clear()` empties the **same** object (every other reference sees an empty list); `lst = []` rebinds only this name to a new list.

Self-check

1. `[1].extend("ab")`? → The list becomes `[1, 'a', 'b']`.
2. Which removal method returns the removed item? → `pop`.
3. What does `remove` do with duplicates? → Removes only the first.

TL;DR

`append` one, `extend` many, `insert` at a position; `remove` by value, `pop` by index (returns it), `del` by index/slice, `clear` everything.

5.5 List methods (complete table) and slicing in depth

The full method table

Method	Does	Cost
`append(x)`	Add `x` at the end	O(1) amortised
`extend(it)`	Add every item of `it`	O(k)
`insert(i, x)`	Insert before index `i`	O(n)
`remove(x)`	Delete first item equal to `x`	O(n)
`pop([i])`	Remove and return item `i` (default last)	O(1) last / O(n)
`clear()`	Remove all items	O(n) (drops each reference)
`index(x[, start[, end]])`	First index of `x`	O(n)
`count(x)`	Number of items equal to `x`	O(n)
`sort(*, key=None, reverse=False)`	Sort in place (stable)	O(n log n)
`reverse()`	Reverse in place	O(n)
`copy()`	Shallow copy	O(n)

Only 11 methods — everything else is a built-in function (`len`, `sorted`, `reversed`, `min`, `max`, `sum`, `enumerate`, `zip`) or an operator.

```
colors = ["red", "blue", "green", "blue"]
print(colors.count("blue"), colors.index("blue"), colors.index("blue", 2))
colors.remove("blue");   print(colors)          # only the first 'blue'
colors.append("amber")
colors.sort();           print("sort:", colors)
colors.reverse();        print("reverse:", colors)
nums = [3, 1, 2]
snapshot = nums.copy()
nums.sort(reverse=True)
print(nums, snapshot)
```

Output:

```
2 1 3
['red', 'green', 'blue']
sort: ['amber', 'blue', 'green', 'red']
reverse: ['red', 'green', 'blue', 'amber']
[3, 2, 1] [3, 1, 2]
```

`index("blue", 2)` starts searching at position 2, so it finds the second `"blue"` at index 3.

Slicing in depth — steps and slice assignment

Everything about string slicing (Chapter 4) applies: `a[start:stop:step]`, stop excluded, negatives from the end, never raises. Because lists are mutable you can also **assign to** and **delete** slices.

Form	Rule
`a[i:j] = iterable`	Replace that range; the new data may be **longer or shorter**
`a[i:i] = iterable`	Empty range → pure **insertion** at `i`
`a[i:j:k] = iterable`	Extended (stepped) slice → lengths **must match**
`del a[i:j:k]`	Delete the selected positions
`a[:] = []`	Empty the list in place (same as `clear()`)

```
x = list(range(10))
print(x[1:8:2], x[::-2], x[-3:])
y = x.copy(); y[2:5] = ["a"];           print("shrink :", y)      # lengths may differ
y = x.copy(); y[2:2] = ["new", "items"]; print("insert :", y)     # empty slice = insert
y = x.copy(); y[::2] = [0] * 5;         print("stepped:", y)     # lengths must match
y = x.copy(); del y[::2];               print("del ::2:", y)
y = x.copy(); y[:] = [];                print("clear  :", y)
try:
    y = x.copy(); y[::2] = [1, 2]
except ValueError as e:
    print("ValueError:", e)
```

Output:

```
[1, 3, 5, 7] [9, 7, 5, 3, 1] [7, 8, 9]
shrink : [0, 1, 'a', 5, 6, 7, 8, 9]
insert : [0, 1, 'new', 'items', 2, 3, 4, 5, 6, 7, 8, 9]
stepped: [0, 1, 0, 3, 0, 5, 0, 7, 0, 9]
del ::2: [1, 3, 5, 7, 9]
clear  : []
ValueError: attempt to assign sequence of size 2 to extended slice of size 5
```

Common mistakes

- Expecting `a[1:3] = 5` to work — the right side must be iterable: `a[1:3] = [5]`.

- Forgetting that a slice **read** is a shallow copy but a slice **assignment** changes the original.

Self-check

1. `x = [0,1,2,3]; x[1:3] = []` → ? → `[0, 3]`.
2. Can `a[::2] = [1]` work on a 4-item list? → No, the stepped slice has 2 slots; lengths must match.
3. How many methods does `list` have? → 11.

TL;DR

11 methods; slicing reads copies, slice assignment edits in place (length may change unless there's a step).

5.6 List comprehensions

What it is

A **list comprehension** builds a new list from an iterable in one expression:

```
[expression for item in iterable if condition]
```

- **expression** – what to put in the new list (e.g. `n ** 2`).
- **for item in iterable** – the source loop.
- **if condition** – optional filter; items that fail it are skipped.

The problem it solves

The traditional pattern – create an empty list, loop, `append` – takes three or four lines to say one thing, and the repeated `.append` attribute lookup and method call cost time. A comprehension says it in one readable line and runs as specialised bytecode.

How it works

1. Python creates an empty list.
2. For each `item` in the iterable, it evaluates `condition` (if present).
3. If true (or absent), it evaluates `expression` and appends the result.
4. Multiple `for` clauses nest **left to right**, in the same order you'd write nested loops.

```
numbers = [1, 2, 3, 4, 5]
squared = []
for n in numbers:                          # the traditional way
    squared.append(n ** 2)
print("loop:", squared)
print("comp:", [n ** 2 for n in numbers])
print("even squares:", [n ** 2 for n in numbers if n % 2 == 0])
print("if/else in expr:", ["even" if n % 2 == 0 else "odd" for n in numbers])
print([f"Number {n} squared is {n ** 2}." for n in numbers[:2]])

matrix = [[1, 2, 3], [4, 5, 6]]
flat = [x for row in matrix for x in row]          # same order as nested for-loops
transposed = [[row[i] for row in matrix] for i in range(3)]
pairs = [(c, n) for c in "ab" for n in (1, 2)]
print(flat, transposed, pairs)

total = sum(n * n for n in range(1_000_000))       # generator expression: no list built
```

Output:

```
loop: [1, 4, 9, 16, 25]
comp: [1, 4, 9, 16, 25]
even squares: [4, 16]
if/else in expr: ['odd', 'even', 'odd', 'even', 'odd']
['Number 1 squared is 1.', 'Number 2 squared is 4.']
[1, 2, 3, 4, 5, 6] [[1, 4], [2, 5], [3, 6]] [('a', 1), ('a', 2), ('b', 1), ('b', 2)]
sum of squares: 333332833333500000
loop+append: 39.0 ms   comprehension: 31.6 ms (ratio 1.2x; machine-dependent)
```

Note the two different `if` positions:

- `[... for n in xs if cond]` – **filter** (fewer items out).
- `[a if cond else b for n in xs]` – **choose a value** (same number of items out).

The speed gain is real but modest (1.2× here). Choose comprehensions for **clarity**; the speed is a bonus.

Trade-offs

	Comprehension	`for` loop + `append`	Generator expression `( … )`	`map` / `filter`
Readability for simple transforms	Best	OK	Best	OK with named functions
Memory	Whole list	Whole list	One item at a time	Lazy iterator
Complex logic, try/except, prints	Poor	Best	Poor	Poor

When not to use it

- When the body needs several statements, error handling or side effects – write a loop.
- When you only iterate once (e.g. feeding `sum`, `any`, `max`, `"".join`) – use a **generator expression** with round brackets to avoid building a list.
- More than two `for` clauses, or nested comprehensions deeper than one level, usually hurt readability (*Effective Python* recommends no more than two control subexpressions).
- For side effects only (`[print(x) for x in xs]`) – that builds a useless list of `None`s.

Common mistakes

- Reversing the order of nested `for`s: `[x for x in row for row in matrix]` fails (`row` not defined yet).
- Expecting the loop variable to leak: in Python 3 comprehension variables are local to the comprehension.

Real-world examples

Data-cleaning code in pandas/ETL scripts (`[c.strip().lower() for c in columns]`), building SQL parameter lists, and flattening API pages (`[item for page in pages for item in page["results"]]`).

In the interview

30-second answer: "`[expr for x in it if cond]` builds a list in one expression; it's clearer and a bit faster than loop-and-append. Use a generator expression when you only need to iterate once, and a normal loop when the logic gets complicated."

- *Q: Order of multiple fors?* A: Left to right, same as nested loops.
- *Q: Dict and set versions?* A: `{k: v for ...}` and `{x for ...}` (Chapter 6).

Self-check

1. `[c.upper() for c in "abc" if c != "b"]`? → `['A', 'C']`.
2. How do you avoid materialising a list in `sum(...)`? → Use a generator expression: `sum(x*x for x in xs)`.
3. Flatten `[[1],[2,3]]`? → `[x for sub in lst for x in sub]` → `[1, 2, 3]`.

TL;DR

`[expr for x in it if cond]`; `if/else` goes before `for`; nested `for`s read left to right; use `( … )` for one-pass use, a loop for complex logic.

5.7 Sorting

What it is

- `lst.sort()` – sorts the list **in place** and returns `None`.
- `sorted(iterable)` – returns a **new** sorted list from any iterable (list, tuple, string, set, dict keys, generator).

Both accept `key=` (a function that computes the value to sort by) and `reverse=True`.

Why it matters

Sorting underlies reports, rankings, deduplication, binary search and "top-k" queries. Knowing `key` functions and **stability** lets you express almost any ordering in one line.

How it works

1. If `key` is given, Python calls it **once per element** and sorts by those results (decorate–sort–undecorate done for you).
2. Elements (or keys) are compared with `<`. Mixed types that can't be compared raise `TypeError`.
3. The algorithm is **Timsort** (since Python 2.3; 3.11 swapped in the "powersort" merge policy). It finds already-ordered runs and merges them: **O(n log n)** worst case, **O(n)** on already-sorted data.
4. It is **stable**: items with equal keys keep their original relative order. That lets you sort by several keys in multiple passes – secondary key first, primary key last.

```
nums = [5, 2, 9, 1]
new = sorted(nums)                  # returns a NEW list, works on any iterable
print(new, nums)
result = nums.sort()                # sorts IN PLACE, returns None
print(nums, result)
print(sorted("banana"), sorted({3, 1, 2}), sorted({"b": 1, "a": 2}))

words = ["banana", "Apple", "cherry", "Date"]
print(sorted(words))                            # uppercase first (code points)
print(sorted(words, key=str.casefold))          # case-insensitive
print(sorted(words, key=len))                   # by length
print(sorted(words, key=len, reverse=True))
people = [("Ann", 31), ("Bob", 25), ("Cid", 31)]
print(sorted(people, key=itemgetter(1)))        # same as key=lambda p: p[1]
```

Output:

```
[1, 2, 5, 9] [5, 2, 9, 1]
[1, 2, 5, 9] None
['a', 'a', 'a', 'b', 'n', 'n'] [1, 2, 3] ['a', 'b']
['Apple', 'Date', 'banana', 'cherry']
['Apple', 'banana', 'cherry', 'Date']
['Date', 'Apple', 'banana', 'cherry']
['banana', 'cherry', 'Apple', 'Date']
[('Bob', 25), ('Ann', 31), ('Cid', 31)]
```

Stability in action: with `key=len`, `"banana"` and `"cherry"` both have length 6, and they come out in their original order (banana before cherry). With `reverse=True` they *still* keep that order – `reverse` flips the comparison, not the tie order, so stability is preserved.

Multi-key sorts — department ascending, then age descending:

```
people = [("Ann", "HR", 31), ("Bob", "IT", 25), ("Cid", "HR", 25), ("Dee", "IT", 31)]
by_age = sorted(people, key=itemgetter(2))
print("by age (ties keep original order):", [p[0] for p in by_age])
print("tuple key :", [p[0] for p in sorted(people, key=lambda p: (p[1], -p[2]))])
tmp = sorted(people, key=itemgetter(2), reverse=True)        # secondary key first
print("two passes:", [p[0] for p in sorted(tmp, key=itemgetter(1))])
```

Output:

```
by age (ties keep original order): ['Bob', 'Cid', 'Ann', 'Dee']
tuple key : ['Ann', 'Cid', 'Dee', 'Bob']
two passes: ['Ann', 'Cid', 'Dee', 'Bob']
```

A **tuple key** compares field by field; negating a number flips its direction. When a field can't be negated (e.g. a string you want descending), use two stable passes instead.

Trade-offs

	`list.sort()`	`sorted()`
Input	Lists only	Any iterable
Result	Same list, changed	New list
Memory	No extra list	Extra O(n)
Keep original?	No	Yes

Common mistakes

- `lst = lst.sort()` → `lst` becomes `None`.
- Sorting mixed types (`[3, "a"]`) → `TypeError: '<' not supported between instances of 'str' and 'int'`.
- Sorting numeric strings as strings (`"10" < "9"`) — use `key=int`.
- Writing a `lambda` when `operator.itemgetter` / `attrgetter` or `str.casefold` already exists (clearer, slightly faster).

Real-world examples

Timsort, designed by Tim Peters for CPython, was adopted for object sorting in Java (`Arrays.sort` on objects, Java 7), on Android, and in V8 (Chrome/Node.js) since 2018. Leaderboards, "sort by price then rating" filters on shopping sites, SQL's `ORDER BY a, b DESC` and pandas' `df.sort_values(["a", "b"], ascending=[True, False])` all map to key-based stable sorts.

In the interview

> **30-second answer:** "`sort` sorts a list in place and returns None; `sorted` returns a new list from any iterable. Both take `key` and `reverse`. Python uses Timsort — stable, O(n log n), O(n) on nearly sorted data — and stability lets you do multi-key sorts with a tuple key or successive passes."

- *Q: Why is stability useful?* A: Equal elements keep prior order, enabling multi-pass sorts (e.g. by name, then by department).
- *Q: How many times is `key` called?* A: Once per element.
- *Q: Top 3 of a million items?* A: `heapq.nlargest(3, items)` — O(n log k) instead of sorting everything (Chapter 6).

Self-check

1. What does `sorted("cab")` return? → `['a', 'b', 'c']` (a list).
2. Sort strings by length, longest first? → `sorted(words, key=len, reverse=True)`.

3. Sort by dept ascending, salary descending? → `key=lambda e: (e.dept, -e.salary)`.

TL;DR

`sort()` in place/None, `sorted()` new list; `key=` once per item; Timsort is stable and adaptive.

5.8 Copying — aliasing, shallow vs deep, and the nested-list trap

What it is

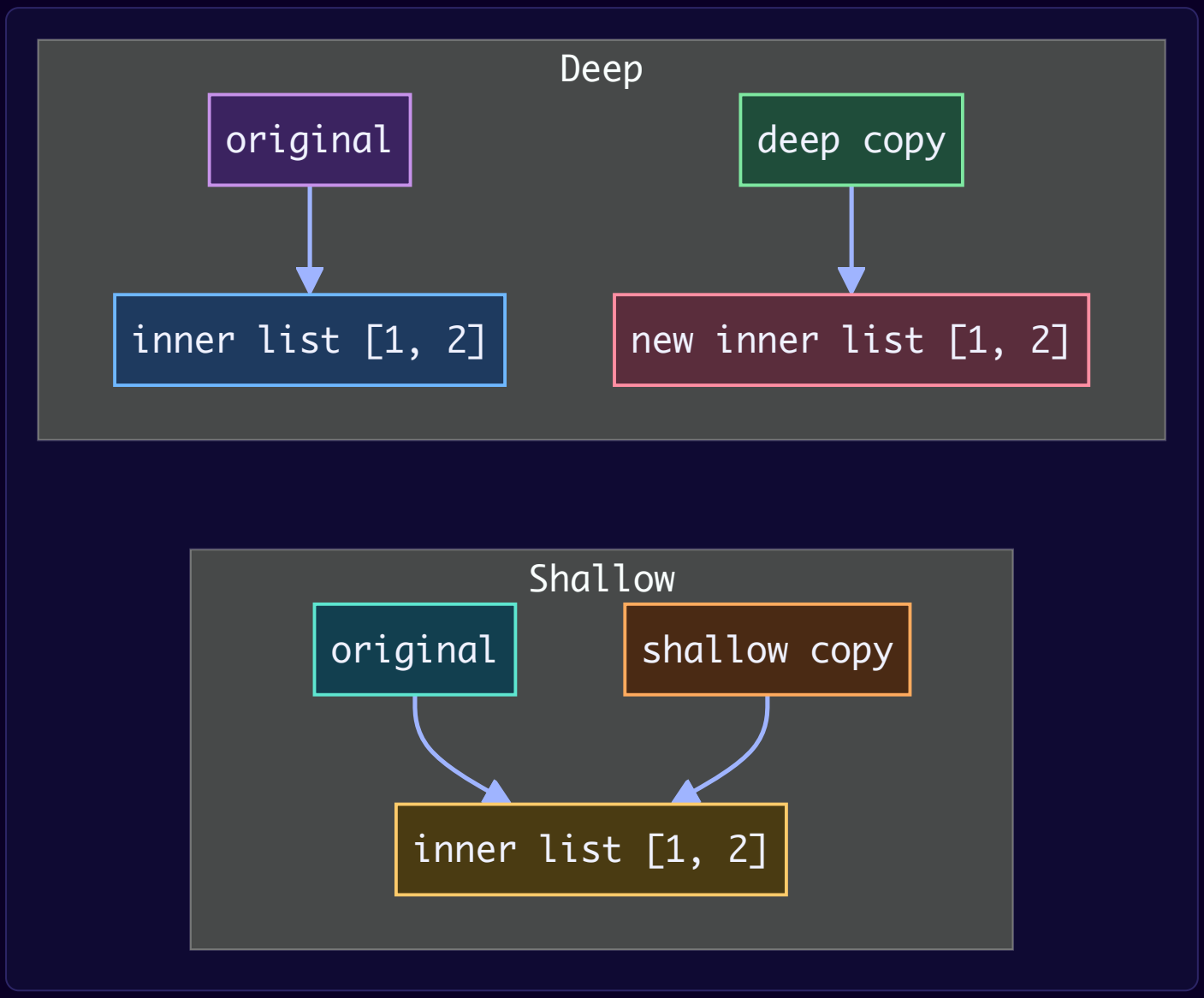
- **Aliasing**: two names referring to the **same** object. `b = a` never copies.
- **Shallow copy**: a new outer container whose slots point to the **same** inner objects. Made by `a.copy()`, `list(a)`, `a[:]`, `copy.copy(a)`.
- **Deep copy**: a new outer container **and** recursive copies of everything inside. Made by `copy.deepcopy(a)`.

Analogy: aliasing is giving a house a second address; a shallow copy is building a new house but sharing the same furniture; a deep copy duplicates the house and every piece of furniture.

Why it matters

Most "my data changed by itself" bugs come from aliasing or shallow copies of nested data.

How it works



```
a = [1, 2, 3]
b = a                       # NOT a copy: two names, one list
b.append(4)
print(a, b, a is b)

original = [[1, 2], [3, 4]]
shallow = original.copy()
deep = copy.deepcopy(original)
original[0].append(99)                  # mutate an INNER list
original.append([5])                    # mutate the OUTER list
print("original:", original)
print("shallow :", shallow, "<- shares inner lists")
print("deep    :", deep, "<- fully independent")
print(shallow[0] is original[0], deep[0] is original[0])
```

Output:

```
[1, 2, 3, 4] [1, 2, 3, 4] True
original: [[1, 2, 99], [3, 4], [5]]
shallow : [[1, 2, 99], [3, 4]] <- shares inner lists
deep    : [[1, 2], [3, 4]] <- fully independent
True False
```

The shallow copy did **not** get `[5]` (its outer list is separate) but **did** see `99` (the inner list is shared).

Passing a list to a function also passes a reference — the function can mutate your list:

```
[1, 2, 3, 4, 'x']       # after add_item(a) appended 'x'
```

The `[[0] * 3] * 3` trap

`*` repeats **references**. `[[0] * 3] * 3` creates one inner row and puts three pointers to it in the outer list.

```
bad = [[0] * 3] * 3                     # 3 references to ONE row
bad[0][0] = 1
good = [[0] * 3 for _ in range(3)]      # 3 separate rows
good[0][0] = 1
```

Output:

```
bad : [[1, 0, 0], [1, 0, 0], [1, 0, 0]] ids: 1 distinct row(s)
good: [[1, 0, 0], [0, 0, 0], [0, 0, 0]] ids: 3 distinct row(s)
```

(`[0] * 3` itself is fine: integers are immutable, so sharing them is harmless.)

The mutable default argument trap

Default values are evaluated **once**, when `def` runs, so a list default is shared by every call:

```
def bad_append(item, bucket=[]):
    bucket.append(item); return bucket
def good_append(item, bucket=None):
    if bucket is None:
        bucket = []
    bucket.append(item); return bucket
print(bad_append(1), bad_append(2))    # [1, 2] [1, 2]
print(good_append(1), good_append(2))  # [1] [2]
```

Trade-offs

	Alias `b = a`	Shallow `a.copy()`	Deep `copy.deepcopy(a)`
Cost	O(1)	O(n)	O(total size of everything)
Independent outer list	No	Yes	Yes
Independent nested objects	No	No	Yes
Handles cycles	–	–	Yes (keeps a memo)

When not to use it

Don't `deepcopy` by reflex – it's slow and copies things you may want shared (caches, connections). If data is flat or contains only immutable items, a shallow copy is a full copy in practice. Consider immutable structures (tuples, frozen dataclasses) so copying isn't needed.

Common mistakes

- `grid = [[None] * cols] * rows` – use a comprehension.
- Thinking `a[:]` copies nested lists – it doesn't.
- `def f(x=[])` / `def f(d={})` – use `None` and create inside.

In the interview

> **30-second answer:** "Assignment never copies – it creates another reference. A shallow copy (`copy()` , `list()` , `[:]`) duplicates only the outer container; nested objects are shared. `copy.deepcopy` copies recursively. Classic traps: `[[0]*3]*3` shares one row, and mutable default arguments are shared between calls."

- *Q: Does `a[:]` copy?* A: Shallowly, yes.
- *Q: Does deepcopy handle cycles?* A: Yes – it tracks already-copied objects in a memo dict.

Self-check

1. `a = [[1]]; b = a[:]; b[0].append(2)` – what is `a` ? → `[[1, 2]]` .
2. Correct way to build a 3×3 zero grid? → `[[0]*3 for _ in range(3)]` .
3. Why is `def f(x=[])` dangerous? → The same list is reused across calls.

TL;DR

`=` aliases, `copy()` shares the insides, `deepcopy` shares nothing. Never multiply a list of lists.

5.9 Tuples – immutable sequences and records

What it is

A **tuple** is an ordered, **immutable** sequence. The **commas** make it; parentheses are just for grouping/readability: `(1, 2, 3)` or `1, 2, 3` (**tuple packing**). A one-element tuple needs a trailing comma: `(42,)` .

Analogy: a list is a shopping list you keep editing; a tuple is a printed receipt – you can read and copy it, not edit it.

Why it matters

- **Integrity:** data that shouldn't change (coordinates, RGB colours, database rows, function return bundles) can't be changed by accident.
- **Hashable** (if all its items are hashable) → can be a dict key or set element: `{(59.9, 10.7): "Oslo"}` .

- **Lighter** than lists: fixed size, no spare capacity. A 3-item tuple is 72 bytes vs 88 for a list on this build. CPython can also build tuples of constants at compile time.
- **Thread safety by construction:** an object nobody can mutate can be shared between threads without locks (the objects inside may still be mutable, see below).

How it works — define, access, "update", loop, operators, slice

```
t = (1, 2, 3, 4, 5)
packed = "apple", "banana", "cherry"          # the commas make the tuple
single, not_a_tuple, empty = (42,), (42), ()
print(t, packed, type(single).__name__, type(not_a_tuple).__name__, empty)
print(t[0], t[2], t[-1], t[1:4], t[::-1])
try:
    t[1] = 20
except TypeError as e:
    print("TypeError:", e)
as_list = list(t); as_list[1] = 20; t2 = tuple(as_list)   # "update" = build a new tuple
print("updated copy:", t2)
for item in t:
    print(item, end=" ")
a, b = (1, 2, 3), (4, 5, 6)
print(a + b, a * 2, 3 in a, 7 not in a, (1, 2) < (1, 3))
```

Output:

```
(1, 2, 3, 4, 5) ('apple', 'banana', 'cherry') tuple int ()
1 3 5 (2, 3, 4) (5, 4, 3, 2, 1)
TypeError: 'tuple' object does not support item assignment
updated copy: (1, 20, 3, 4, 5)
1 2 3 4 5
(1, 2, 3, 4, 5, 6) (1, 2, 3, 1, 2, 3) True True True
```

Operator	Meaning	Example
`+`	Concatenate into a new tuple	`(1, 2) + (3,)` → `(1, 2, 3)`
`*`	Repeat	`(1, 2) * 2` → `(1, 2, 1, 2)`
`in` / `not in`	Membership	`3 in (1, 2, 3)` → `True`
`<`, `==`, …	Lexicographic comparison	`(1, 2) < (1, 3)` → `True`
`[i]`, `[i:j:k]`	Index and slice (slices are new tuples)	`(0,1,2,3)[1:3]` → `(1, 2)`

To "update" a tuple: convert to a list, change it, convert back — or build a new tuple with slicing: `t[:1] + (20,) + t[2:]`.

Immutability is shallow

The tuple's **slots** can't change, but a mutable object **inside** can:

```
t = ([1, 2], "x")
t[0].append(3)
print(t)                  # ([1, 2, 3], 'x')
hash(t)                   # TypeError: unhashable type: 'list'
```

A tuple is hashable only if everything inside is hashable.

Tuples as records

Think of a tuple as a **record** where *position has meaning*: `("Ann", "HR", 31)` = name, department, age. Lists are for **many things of the same kind** (a list of employees); tuples are for **one thing with several fields** (one

employee). That's the guidance in *Fluent Python*, and it's why functions return tuples: `divmod(17, 5)` → `(3, 2)`.

Common mistakes

- `(42)` is just the int 42 — write `(42,)`.
- Trailing comma accidents: `x = 1,` makes `x` a tuple.
- Assuming a tuple of lists is safe to use as a dict key — it isn't hashable.

Self-check

1. `type(("a"))`? → `str`. `type(("a",))`? → `tuple`.
2. Can you append to a list inside a tuple? → Yes; the tuple still holds the same list object.
3. Why can tuples be dict keys and lists cannot? → Tuples (of hashable items) are immutable and hashable.

TL;DR

Commas make tuples; they're immutable (shallowly), hashable if their contents are, lighter than lists, and ideal as fixed-shape records.

5.10 Unpacking tuples (and any iterable)

What it is

Unpacking assigns the items of a sequence to several names in one statement: `a, b, c = (1, "Hello", 3.14)`. It works with any iterable — lists, strings, generators, dict items.

Why it matters

It removes index noise (`row[0]`, `row[1]`) and so removes off-by-one bugs, and it's how you receive multiple return values, swap variables and loop over pairs.

How it works

1. The right side is evaluated fully first.
2. Python iterates it and binds each item to the next name on the left.
3. The counts must match — otherwise `ValueError` ("not enough values" / "too many values").
4. **Star-unpacking** (`*name`) collects "everything else" into a **list** (possibly empty). Only one star per assignment.
5. Unpacking nests: `for name, (lat, lon) in ...`.

```
a, b, c = (1, "Hello", 3.14)
print(a, b, c)
try:
    w, x, y, z = (1, "Hello", 3.14)
except ValueError as e:
    print("ValueError:", e)
first, *middle, last = (1, 2, 3, 4, 5, 6)
print(first, middle, last)
head, *tail = [10]
print(head, tail)
x, y = 1, 2
x, y = y, x                                  # swap
print(x, y)
for name, (lat, lon) in [("Oslo", (59.9, 10.7)), ("Lima", (-12.0, -77.0))]:
    print(f"{name}: {lat}, {lon}")
_, _, ext = "archive.tar.gz".partition(".")   # _ = "don't care"
print(ext)
```

Output:

```
1 Hello 3.14
ValueError: not enough values to unpack (expected 4, got 3)
1 [2, 3, 4, 5] 6
10 []
2 1
Oslo: 59.9, 10.7
Lima: -12.0, -77.0
tar.gz
```

Wrapping risky unpacking in `try/except ValueError` (as above) lets a program report bad input (say, a malformed CSV row) instead of crashing.

The swap `x, y = y, x` works because the right side builds the tuple `(2, 1)` **before** any assignment happens.

Star in calls and literals (the same idea, in reverse):

Syntax	Meaning	Example
`f(*seq)`	Spread items as positional arguments	`print(*[1, 2, 3])` → `1 2 3`
`[*a, *b]`	Build a list from several iterables	`[*(1, 2), *[3]]` → `[1, 2, 3]`
`(*a, x)`	Build a tuple	`(*t, 99)`
`def f(*args)`	Collect extra positional arguments into a tuple	Chapter on functions

Common mistakes

- Using two stars: `a, *b, *c = ...` → `SyntaxError`.
- Expecting `*middle` to be a tuple – it is always a **list**.
- Unpacking a dict directly gives keys; use `.items()` for pairs.

In the interview

> **30-second answer:** "Unpacking binds items of any iterable to names; the count must match or you get ValueError. `first, *rest = seq` gathers the leftovers into a list. It powers multiple return values, swaps without a temp, and clean loops like `for k, v in d.items()`."

- *Q: Is `a, b = b, a` atomic?* A: It's evaluated as build tuple, then assign – no temp variable needed, but it isn't a thread-atomic operation.
- *Q: What does `*_` do?* A: Discards any number of items: `first, *_ = row`.

Self-check

1. `a, *b = "hey"` ? → `a='h'` , `b=['e', 'y']`.
2. `x, y = [1, 2, 3]` ? → `ValueError: too many values to unpack (expected 2)`.
3. How to swap `a` and `b` ? → `a, b = b, a`.

TL;DR

`a, b, c = seq`; counts must match; `*rest` gathers a list; `_` for ignored slots; works in `for` loops and nests.

5.11 Joining tuples

What it is

Combining several tuples into one. Because tuples are immutable, every join creates a **new** tuple.

How it works — five ways

```
parts = [(1, 2), (3, 4), (5, 6)]
out = ()
for p in parts:
    out += p                                # new tuple every time: O(n^2) overall
print("loop     :", out)
print("+        :", (1, 2, 3) + (4, 5, 6))
print("sum      :", sum(parts, ()))          # also O(n^2): repeated +
print("genexpr  :", tuple(x for p in parts for x in p))
print("chain    :", tuple(itertools.chain.from_iterable(parts)))
```

Output (last line: joining 3,000 two-item tuples):

```
loop     : (1, 2, 3, 4, 5, 6)
+        : (1, 2, 3, 4, 5, 6)
sum      : (1, 2, 3, 4, 5, 6)
genexpr  : (1, 2, 3, 4, 5, 6)
chain    : (1, 2, 3, 4, 5, 6)
3000 tuples: sum 14.55 ms vs chain 0.052 ms
```

- `+` – perfect for two or three tuples.
- `+=` in a loop and `sum(parts, ())` – both repeatedly create a bigger tuple and copy everything so far, so joining m tuples costs $O(\text{total}^2)$. `sum` is **not** faster than the loop; it does the same work. (The docs recommend `itertools.chain` for concatenating sequences – `sum` even refuses strings outright.)
- `tuple(x for p in parts for x in p)` – a flattening generator expression: the outer `for` walks each tuple, the inner `for` walks its items, and `tuple()` collects them once → O(total).
- `tuple(itertools.chain.from_iterable(parts))` – the idiomatic linear solution, implemented in C: ~280× faster here.

Common mistakes

- Joining a tuple and a list: `(1,) + [2]` → `TypeError`. Convert first.
- Forgetting the comma when adding one item: `t + (4)` fails; `t + (4,)` works.

Self-check

1. Fastest general way to join many tuples? → `tuple(itertools.chain.from_iterable(tuples))`.
2. Is `sum(list_of_tuples, ())` linear? → No, quadratic.

TL;DR

Few tuples → `+`. Many → `tuple(chain.from_iterable(...))` or a flattening generator. Avoid `+=` loops and `sum`.

5.12 Tuple methods

Tuples have only two methods – everything that would modify is missing by design.

Method	Returns	Cost
`count(x)`	How many items equal `x`	O(n)
`index(x[, start[, end]])`	Position of the first `x`; `ValueError` if absent	O(n)

```
fruit = ("apple", "banana", "cherry", "apple", "cherry")
print(fruit.count("apple"), fruit.count("cherry"), fruit.index("banana"), fruit.index("cherry"))
```

Output:

```
2 2 1 2
```

`index("cherry")` returns `2` – the **first** cherry – even though another is at 4. The built-ins `len`, `min`, `max`, `sum`, `sorted` (returns a list), `reversed`, `any`, `all` all work on tuples too.

TL;DR

`count` and `index`, both O(n). Everything else comes from built-ins.

5.13 `namedtuple` and `typing.NamedTuple` – tuples with field names

What it is

`collections.namedtuple` creates a **tuple subclass** whose positions also have names: `p.x` instead of `p[0]`. `typing.NamedTuple` is the same thing written as a class with type hints and defaults.

The problem it solves

`emp[2]` says nothing; `emp.salary` says everything. Named tuples keep all tuple benefits (immutable, hashable, unpackable, as small as a tuple – no per-instance `__dict__`) while making code self-documenting.

How it works

```
Point = namedtuple("Point", ["x", "y"])
p = Point(3, 4)
print(p, p.x, p[1], p._asdict(), p._replace(x=10))
x, y = p                                    # still a tuple

class Employee(NamedTuple):                 # typed, class-style syntax
    name: str
    dept: str
    salary: int = 0
e = Employee("Ann", "HR")
print(e, e.salary)
```

Output:

```
Point(x=3, y=4) 3 4 {'x': 3, 'y': 4} Point(x=10, y=4)
Employee(name='Ann', dept='HR', salary=0) 0
```

Useful extras: `_asdict()` (to a dict, e.g. for JSON), `_replace(**changes)` (a modified **copy**), `_fields` (field names), `_make(iterable)` (build from a row).

Trade-offs

	tuple	namedtuple / NamedTuple	@dataclass	dict
Access by name	No	Yes	Yes	Yes (d["k"])
Immutable	Yes	Yes	Optional (frozen=True)	No
Hashable	If items are	If items are	If frozen	No
Memory	Smallest	Same as tuple	Bigger (use slots=True to shrink)	Biggest
Methods / validation	No	Limited	Full class	No
Also behaves as a sequence	Yes	Yes (can surprise: p == (3, 4) is True)	No	No

When not to use it

When records need mutation, validation or methods, prefer `@dataclass`. When equality with plain tuples would be a bug, prefer a dataclass too.

Real-world examples

The standard library returns named tuples from many functions: `urllib.parse.urlparse()` → `ParseResult(scheme=..., netloc=...)` (a namedtuple subclass). `sys.version_info` (`sys.version_info.major`), `time.struct_time` and `os.stat_result` are *struct sequences* – the C-level cousin with the same "tuple plus named fields" behaviour. Database drivers use them for rows too (e.g. psycopg 3's `namedtuple_row` row factory).

In the interview

> **30-second answer:** "A namedtuple is a tuple subclass with named fields – immutable, lightweight, unpackable, with `_asdict` and `_replace`. I use it (or `typing.NamedTuple` for type hints) for simple immutable records and switch to dataclasses when I need mutability, validation or methods."

Self-check

1. How do you "change" a field? → `p2 = p._replace(x=10)` – returns a new object.
2. Is a namedtuple equal to a plain tuple with the same values? → Yes, which can be surprising.

TL;DR

Named fields on a tuple. Use for immutable records; use dataclasses for richer objects.

5.14 Choosing: `list` vs `tuple` vs `array.array` vs NumPy

What it is

Four sequence types, four sweet spots. NumPy's `ndarray` (third-party, `pip install numpy`) is a typed, contiguous, possibly multi-dimensional array whose operations run in compiled loops over the whole array at once (**vectorisation**).

How they compare – measured

One million integers, "multiply every element by 2":

```
container         memory    x*2 time
list              36.0MB      20.8ms
array.array        8.2MB      45.7ms
numpy              8.0MB       0.6ms
```

- `list` stores pointers + int objects → most memory; element-wise work runs in the interpreter.
- `array.array('q')` stores raw 8-byte ints → compact, but every read creates a Python int, so a Python-level loop is **slower** than the list.
- NumPy stores raw 8-byte ints **and** does the arithmetic in C for the whole array → ~35× faster than the list here.

Trade-offs

	`list`	`tuple`	`array.array`	`numpy.ndarray`
Mutable	Yes	No	Yes	Yes (fixed size)
Element types	Any, mixed	Any, mixed	One C numeric type	One dtype (numeric, bool, fixed strings, objects)
Memory for numbers	High	High	Low	Low
Hashable	No	Yes (if items are)	No	No
Fast element-wise maths	No	No	No	Yes (vectorised)
Multi-dimensional	Via nesting	Via nesting	No	Yes
Standard library	Yes	Yes	Yes	No (third-party)
`+` means	Concatenate	Concatenate	Concatenate	**Element-wise add**

When to use which

- **`list`** – default for general collections that change.
- **`tuple`** – fixed records, dict keys, return values, constants.
- **`array.array`** – lots of numbers kept compactly with no dependency; binary file/network formats.
- **NumPy** – any numerical computing: statistics, images, ML features, linear algebra. It's the foundation under pandas, scikit-learn, SciPy and matplotlib.

Common mistakes

- Looping over a NumPy array in Python (`for x in arr: ...`) – you lose the speed. Use vectorised expressions (`arr * 2`, `arr[arr > 0]`).
- Expecting `np_a + np_b` to concatenate – it adds element-wise (use `np.concatenate`).
- Picking `array.array` for speed – it's for memory.

In the interview

30-second answer: "List for general mutable data, tuple for immutable records and dict keys, `array.array` for compact homogeneous numbers without dependencies, NumPy for fast numerical work because it stores raw values contiguously and runs vectorised C loops."

Self-check

1. Why is `array.array` slower than a list for `[x*2 for x in arr]`? → Every element read must create a Python int object.
2. What does `+` do on NumPy arrays? → Element-wise addition.
3. Which of the four can be a dict key? → Tuple (if its items are hashable).

TL;DR

General → list. Fixed → tuple. Compact numbers → array. Fast numbers → NumPy.

Chapter TL;DR

Topic	Remember
`array` module	Typed, compact, mutable; type codes set size/range; no `sort` / `copy` methods; `clear()` from 3.13; memory win, not speed win.
Lists	Ordered, mutable, mixed types; index/slice/update; `+` new, `extend` in place.
Internals	Dynamic array of pointers; ~12.5% over-allocation → amortised O(1) append; front/middle insert/delete O(n); `in` O(n).
Add/remove	`append`, `extend`, `insert`; `remove` (value), `pop` (index, returns), `del`, `clear`.
Slicing	`[start:stop:step]`; slice assignment can resize; stepped assignment needs equal length.
Comprehensions	`[expr for x in it if cond]`; generator expression for one-pass use; loops for complex logic.
Sorting	`sort()` in place → None; `sorted()` new list; `key`, `reverse`; Timsort is stable, O(n log n), O(n) on sorted input.
Copying	`=` aliases; `copy()` / `[:]` shallow; `deepcopy` recursive; never `[[0]*n]*m`; no mutable defaults.
Tuples	Commas make them; `(x,)` for one; shallowly immutable; hashable if contents are; records.
Unpacking	Counts must match; `*rest` → list; swap with `a, b = b, a`.
Joining tuples	`+` for few; `chain.from_iterable` for many; `sum` / `+=` loops are quadratic.
Tuple methods	`count`, `index`.
namedtuple	Named, immutable records; `_asdict`, `_replace`; dataclass for more.
Choosing	list / tuple / array / NumPy by mutability, memory, and speed needs.